MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY 11748240

(SIGHT CURRENT GENERATOR ADAPTER)

This specification is approved for use **by all Departments and Agencies of the Department of Defense,**

1. SCOPE

1.1 This specification covers the requirements for a Sight Current Generator Adapter circuit board, herein referred to as the "SCG Adapter" which forms a part of Electronic System Test Set AN/TSM 100,

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form *a* **part of this specification to the extent specified herein.**

SPECIFICATIONS

Military

MIL-T-5422	Testing, Environmental, Aircraft **Electronic Equipment**
MIL-F-13926	**Fire Control Materiel: General Specifications Governing the Manufacture and Inspection of**
MIL-P-14232	**Parts, Equipment and Tools for Army Materiel:** Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction; M18 Manufacturing processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and **Tables for Inspection** by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

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D11748242	Schematic, SCG Adapter Board
D11748240	Circuit Card Assembly, SCG Adapter
C11739937-3	Receptacle

PACKAGING DATA SHEET

D11748240	**Packaging of Circuit Card Assembly, SCG Adapter**

(Copies of specifications, standards, drawings, and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer,)

3. REQUIREMENTS

3.1 Fabrication.- The SCG Adapter shall be manufactured in accordance with Drawing D11748240 and drawings pertaining thereto.

3.2 General specifications.- Where applicable, the assemblies shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2,1 MIL-F-13926.

a, Order of precedence.

b. Dimensions and tolerances.

c. Inorganic protective surface finishes.

d. Part identification and marking.

e. Workmanship.

f. Soldering electrical connections.

3.2.2 MIL-C-46357.

a. Etched circuit assembly.- The requirements for etched circuit assemblies of MIL-C-46357 shall apply with modifications as follows:

(1) **Plated etched circuit boards.- In addition, assemblies shall not contain copper or nickel visible through hot solder dip, when viewed at 10 times magnification,**

(2) Locating grid.- The component mounting holes shall be within 0.007 inch of the true grid intersection, instead of 0.005 inch,

(3) Drying.- The drying process shall not be required.

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3.3 Environmental service conditions,

3.3.1 Vibration.- The SCG Adapter board shall be mounted on the vibration table in normal service position. The SCG Adapter board shall meet the requirements of 3.4 and 3.5 after being subjected to a simple harmonic motion having excursions as follows:

Frequency	Excursion (Double Amplitude)
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 55	**0.02 inch**

The frequency shall be varied logarithmically between 5 and 55' cycles per second (cps) within 15 minutes and returned to 5 cps in 15 minutes. The motion shall be applied to the assemblies for a period of 30 minutes in each direction of the 3 mutually perpendicular axes (total of 90 minutes), relative to its normal operational position. Vibration tests shall be conducted under ambient room conditions.

, 3.3.2 Temperatures.

3.3.2.1 Operating,- The SCG Adapter board shall meet the requirements of 3.4 and 3.5 while exposed to temperature of +32°F and +1.31°F.

3.3.2.2 Storage.- The SCG Adapter board shall meet the requirements of 3,4 and 3.5 at a temperature of 60°F to 90°F after being exposed to each temperature of -80° F and +185° F.

3.3.3 Humidity.- The SCG Adapter board shall meet the requirements of 3.4 and 3.5 after being subjected to ten consecutive humidity cycles at. relative humidities between 85 and 95 percent and temperatures from 82°F to 160°F **over a period not less than 240 hours.**

3.3.4 Shock,- The SCG Adapter board shall meet the requirements of 3.4 and 3.5 after being subjected to a total of 18 shocks consisting of three shocks in opposing directions along each of three mutually perpendicular axes. Each shock shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 ± 1 millisecond.

3.3.5 Altitude.

3.3.5.1 Operating.- The SCG Adapter board shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.5.2 Non-operating.- The SCG Adapter board shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressures encountered at 40,000 feet above sea level (5.5 inches Hg) .

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3.3.6 Sand and dust.- The SCG Adapter board shall meet the requirements of 3.4 and 3.5, in both operating and non-operating condition, during and after exposure to sand and dust particles as encountered in desert area.

3.3.7 Salt spray.- The SCG Adapter board shall meet the requirements of 3.4 and 3.5, in both operating and non-operating condition, during and after exposure to a salt-sea atmosphere.

3.4 Electrical.

3.4.1 Continuity.- Each conductor shall be continuous electrically as specified on the schematic diagram D11748242.

3.4.2 Resistance.- Resistance shall be 22 ohms ± 5% when measured between the following terminals:

From	To
P3-E	P3-B
P3-G	P3-C

3.5 Performance.

3.5.1 Conditions of operation.- Unless otherwise specified, the requirements of 3.5.2 shall be met with the input power connections listed in Table I.

TABLE I - INPUT VOLTAGES

Connection	Voltage
P3-G	+15 Vdc ± 2.0 Percent
P3-E	-15 Vdc ± 2.0 percent
P3-A	Return

3.5.2 Output levels.

3.5,2.1 Pin B output.- No load output shall be -15.0 ± 0.3 Vdc, between pin B (-) and pin A (+).

3.5.2..2 Pin C output.- No load output shall be +15.0 ± 0.3 Vdc, between pin C(+) and pin A(-).

3.5.2.3 Pin J output.- Variations of R3 shall provide output voltages between -6.8 and -7.2 Vdc inclusive, when measured between pin J(-) and pin A(+), When R3 is adjusted to provide -7.0 ± .01 percent, this *voltage* shall not vary more than ± 0.1% during exposure to temperature extremes of +32°F to +130°F.

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3.5.2.4 Pin K output.- The output measured between pin K and pin A, shall be a triangular waveform, with an amplitude from -8.47 ± 1.03V to +8.47 ± 1.03v with a period of 0.8 ± 0,11 milliseconds.

3.5.2.5 Pin H output.- The output measured between pin H and pin A, shall be a triangular waveform, with an amplitude from -8.47 ± 1.03V to +8.47 ± 1.03V with a period of 1.0 ± 0.11 milliseconds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General Provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial Production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three SCG Adapter Boards manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior *to* submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample . and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

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4.3.1 Submission of product.- **Unless otherwise specified by the** Contracting Officer, inspection lot **size, lot information, and presenta-** tion of lots shall be in accordance **with "Submission of Product"** provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in **MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables 11 and 111 herein, and test methods and procedures in 4.6. Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection** by lot.

TABLE 11 - CLASSIFICATION OF DEFECTS

(100% INSPECTION)

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Continuity	3.4.1	4.6.2.1
102.	Resistance	3.4.2.	4.6.2.2
103.	Pin B output	3.5,2.1	4.6.3.2
104.	Pin C output	3.5,2.2	4.6.3.3
105.	Pin J output	3.5,2.3	4.6.3.4
106.	Pin K output	3.5.2.4	4.6.3.5
107,	Pin H output	3.5.2.5	4.6.3.6

4.3.3 Acceptance or rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- One SCG **Adapter board out of the first 25** produced and one out of each **25 produced thereafter shall be selected** at random. The sample shall **have met the requirements and tests in** Table II and shall then meet **the requirements and tests in Table III.**

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TABLE III - SPECIAL SAMPLING

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1	Applicable dwg. - Visual
302. General specifications	3.2	Visual - Tactile
303. Vibration	3.3.1	4.6.1.1
304, Temperature	3.3.2	4.6.1.2
305. Humidity	3.3.3	4.6.1.3
306. Shock	3*3*4	4.6.1.4
307 . Altitude	3.3.5	4.6.1.5
"308. Sand and dust	3.3.6	4.6.1.6
309 . Salt spray	3*3.7	4.6.1.7

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment.- Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Contractor design.- The contractor shall design and supply inspection equipment compatible for "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is considered normally to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to ensure that the test equipment will indicate reliable acceptability of a product that does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of the test equipment.

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4.5.2..2 Test equipment.- In conjunction with 4,5.2.1, the test equipment found in Table IV shall be used in performance of the applicable tests as specified in 4.6.

TABLE IV - TEST. EQUIPMENT LIST

ITEM	EQUIPMENT	DESCRIPTION
1.	Power Supply	15 ± 0.1 vdc, regulated, AC ripple: Less than 100 MV peak-to-peak, Capacity: 100 ma or greater.
2.	Power Supply	-15 ± 0.1 Vdc, regulated. AC ripple: Less than 100 MV peak-to-Peak. Capacity: 100 ma or greater.
3.	Switches	Two - SPST.
4.	Oscilloscope	Voltages and time accuracy and resolution of 5.0 percent or better. Rise time: 100 ns or better. Voltage Sensitivity: O.O5V/cm to 10.0 V/cm.
5.	Digital Voltmeter	Accuracy and resolution of .01 percent or better from zero to 100V, Input impedence: 10 megohms or greater on all scales.
6.	Ohmmeter	Accuracy and resolution of one percent from zero to 100 ohms.

4.6 Test methods and procedures,

4.6,1 Environmental.- Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and apparatus requirements of MIL-STD-810.

4.6.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitude and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements of 3.3.1, Subsequent to the vibration test, the SCG adapter shall meet the performance requirements of 3.4 and 3.5,

4.6.1.2 Temperature.- This test shall be performed as specified in Procedure I of Method 501 in MIL-STD-810 for high temperature operating and non-operating conditions as specified in 3.3.2. Low temperature, operating and non-operating, conditions shall be in accordance with

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Procedure I of Method 502 under the conditions specified in 3.3.2.

4.6.1.3 Humidity.- The humidity test shall be performed in accordance with Procedure I of Method 507 in MIL-STD-810. Subsequent to the humidity test, the assembly shall be tested to determine compliance with 3.3.3.

4.6.1.4 Shock.- The shock test shall be performed in accordance with MIL-T-5422.

4.6.1.5 Altitude.- Non-operating and operating. This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in step 2 the altitude shall be limited to 40,000 ft. (5.5 in. Hg). Operation of step 3 shall be the performance tests of 4.6.3.

4.6.1.6 Sand and dust.- The sand and dust test shall be performed **in accordance with MIL-T-5422.**

4.6,1.7 Salt spray.- The salt spray test shall be performed in **accordance with MIL-T-5422.**

4.6.2 Electrical.

4.6.2.1 Continuity.- Continuity shall be inspected by standard measuring equipment (WE) and visual examination for compliance to 3.4.1.

4.6.2.2 Resistance.- Resistance shall be measured with an ohmmeter, item 6 of Table IV, for compliance to 3.4.2.

4.6.3 Performance.

4.6.3.1 Conditions of operation.- Power supplies, items 1 and 2 of Table IV, and "switches, item 3 of Table IV, shall be connected to the assembly under test in accordance with Figure 1, using receptacle 11739937-3.

4.6.3.2 Pin B output.- With switch S2 closed the voltage between pin B(-) and pin A(+) measured with a digital, voltmeter, item 5 of Table IV, shall be -15 ± 0.3 Vdc.

4.6.3.3 Pin C output.- With switch S1 **closed,** the voltage between pin C(+) and pin A(-) measured with a digital voltmeter, item 5 of Table IV, shall be +15 ± 0.3 Vdc.

4.6.3.4 Pin J output.- With switches S1 and S2 closed, and the digital voltmeter connected between pin J(-) and pin A(+), R3 shall be adjusted to the extremes of rotation. The voltmeter shall be checked for the presence of the voltage limits of 3.5.2.3. R3 shall then be adjusted to a voltage indication of -7.0 ± .07 Vdc. Without further adjustment the voltage shall not vary more than ± 0.1% when operating temperature testing is performed per 4.6.1.2.

4.6.3.5 Pin K output.- With switches S1 and S2 closed, and the

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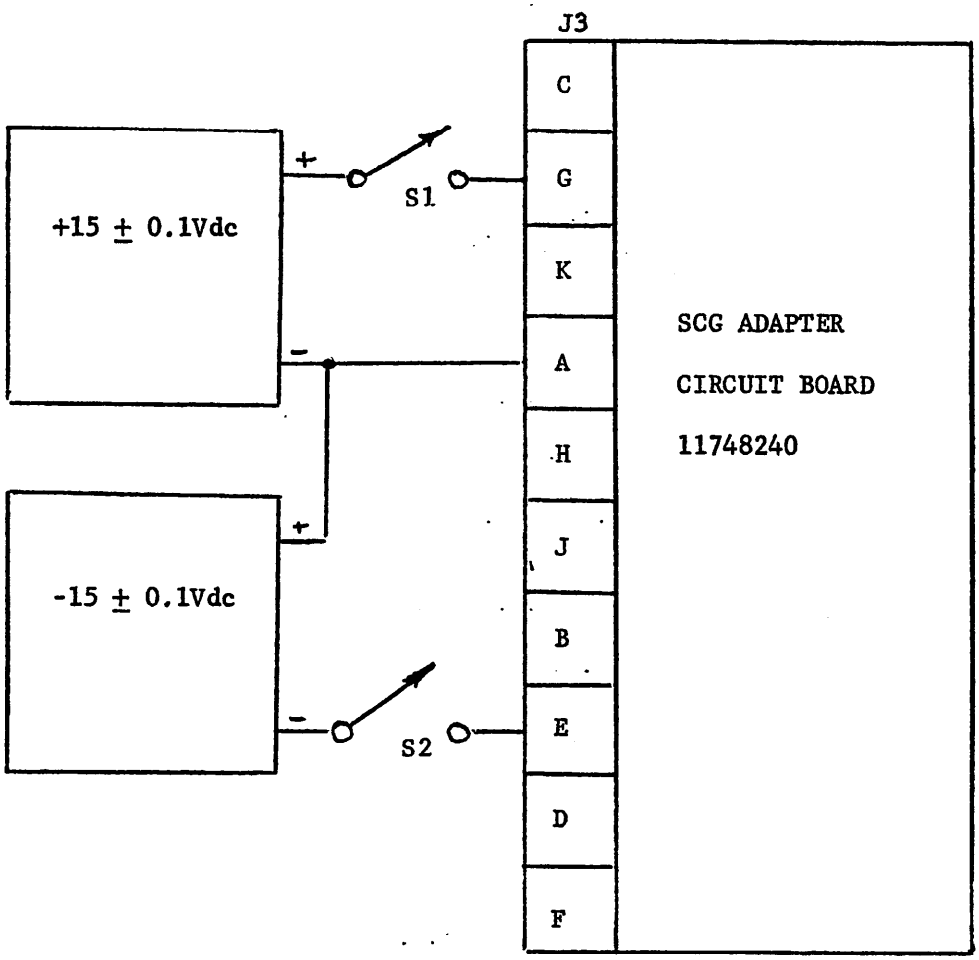


Figure 1 - Test Connections

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oscilloscope, item **4 of Table IV, connected between pin K(+) and Pin A(-) the output of pin K shall be checked for the presence of a triangular waveform. This waveform shall meet the amplitude and period requirements of 3.5.2,4.**

4.6.3.6 Pin H output.- With the switches S1 and S2 closed, and the oscilloscope, item 4 of Table IV, connected between pin H(+) and pin A(-), the output of pin H shall be checked for the presence of a triangular waveform. This waveform shall meet the amplitude and period requirements of 3.5.2.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- packaging and marking shall be in accordance with the level A requirements of Packaging Data Sheet (PDS), AMC Form **1029,** 11748240.

6. NOTES

6.1 The SCG Adapter is used to simulate signals encountered within the Sight Current Generator, in order to test certain subassemblies of that item.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet (see 5.1).

c. Selection of applicable level of preservation, packaging, and packing.

d. Level of protection required.

e. Applicable stock number.

f. Block of serial numbers to identify each SCG Adapter.

Custodian:

Army - MU

Preparing Activity:

Army-MU

Project No. 1285-A436

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE MIL-C-48585(MU) Circuit Card Assembly 11748240 (Sight Current Generator Adapter)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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